



WindSight Group
DAB 26

US WIND MARKET ENTRY STRATEGY

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AGENDA

1 Problem Statement

2 Objectives

3 Approach

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PROBLEM STATEMENT

As of **2022**, **8.9%** of the observed U.S. wind fleet with a recorded commissioning year (**6,252 turbines**) was **at least 20 years old**. As WOW! evaluates market entry, these **aging assets** represent a **potential repowering opportunity**, particularly where older fleets overlap with **high turbine concentrations**.

OBJECTIVES

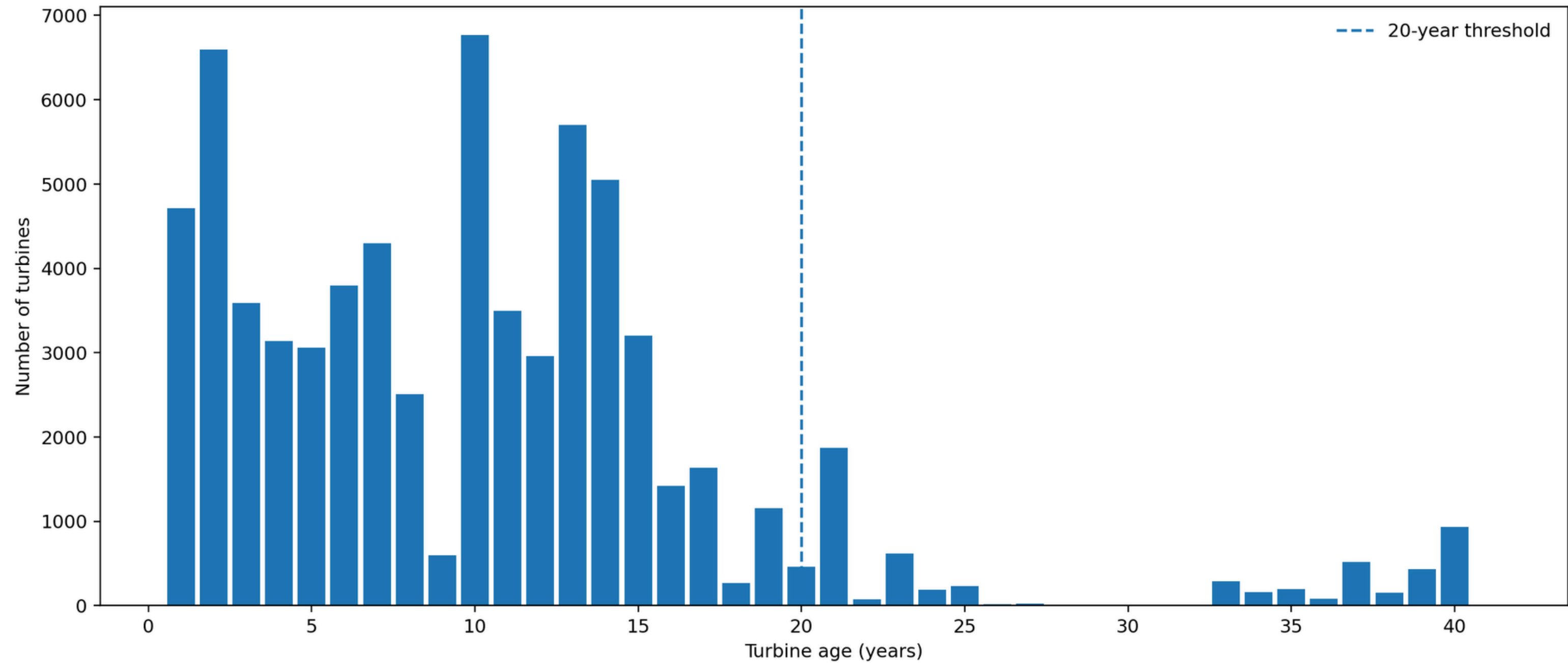
- How old are the observed U.S. wind turbine fleets?
- What portion of the fleet is at least 20 years old?
- How has turbine rated capacity changed over time?
- How does modern turbine capacity compare with older models?
- Which operators have the largest aging turbine fleets?
- Where are aging turbines most concentrated?

APPROACH

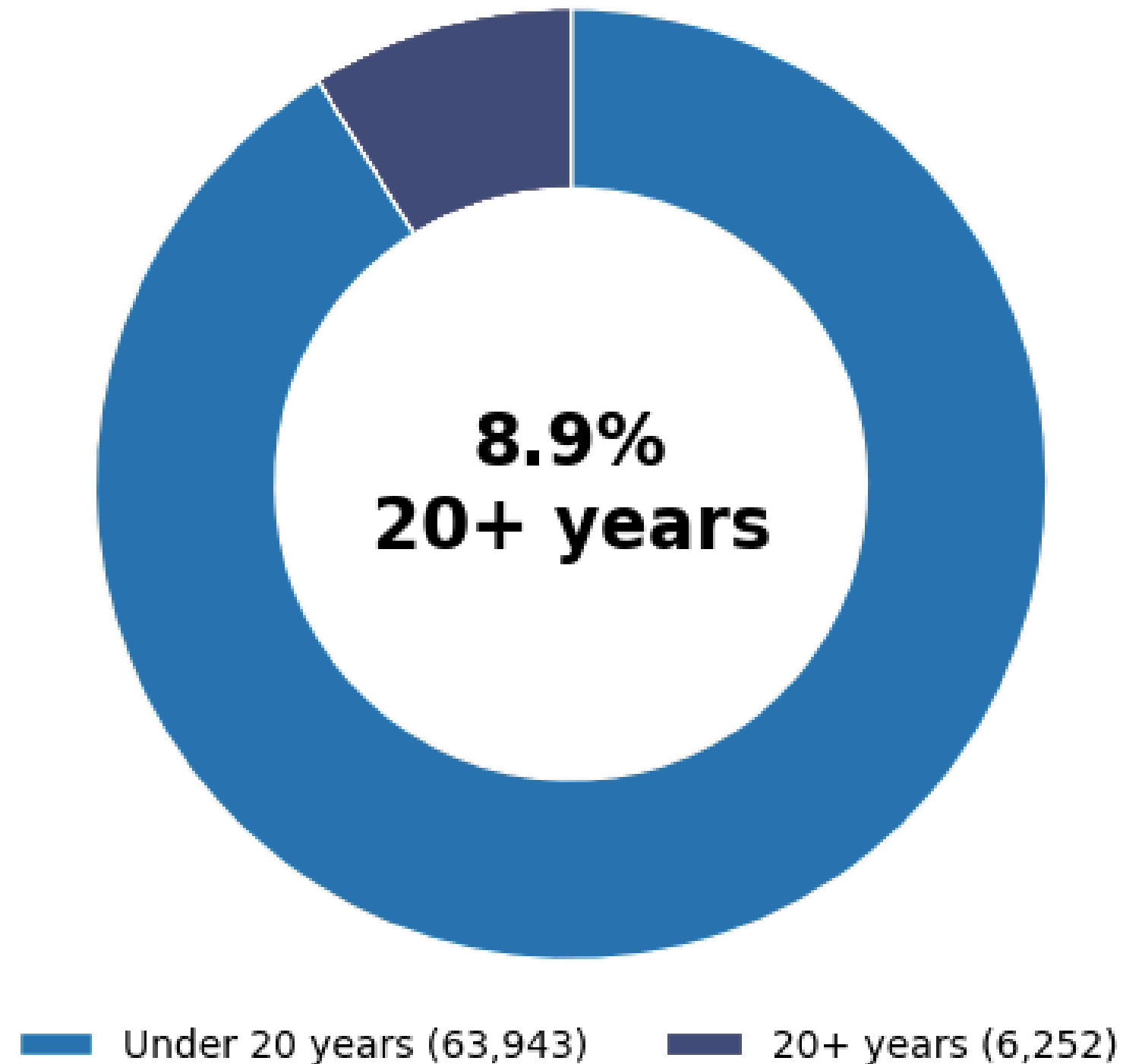
- **Data Preparation:** Cleaned and validated U.S. wind turbine data and EIA 923 operational records.
- **Fleet Age Profiling:** Calculated turbine age as of 2022 and identified turbines aged 20 years or more.
- **Capacity Analysis:** Examined changes in turbine rated capacity across commissioning years.
- **Operator Screening:** Linked aging turbines to EIA 923 operator records to identify operators with the largest matched aging fleets.

HOW OLD ARE THE FLEETS?

Turbine Age Distribution as of 2022

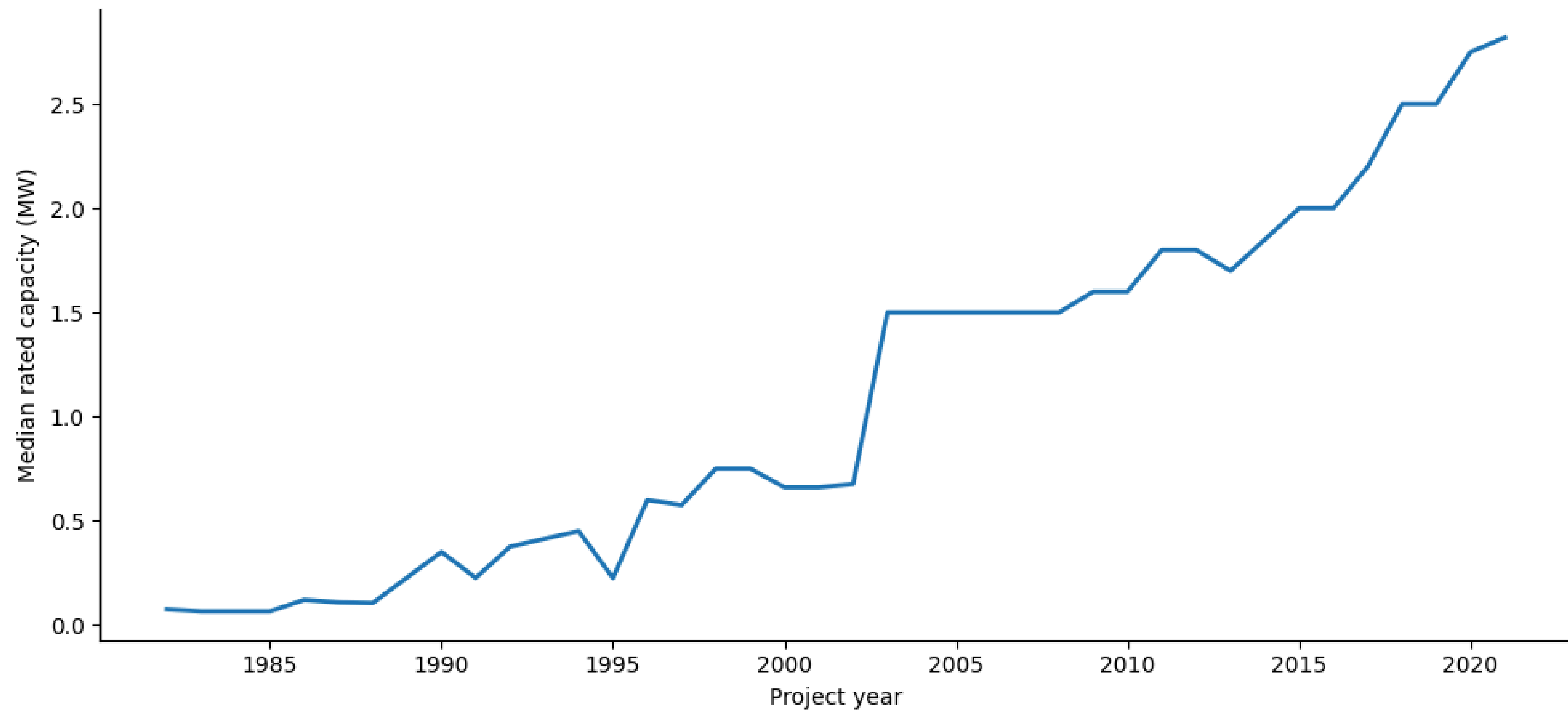


WHAT PORTION OF THE FLEET IS AT LEAST 20 YEARS OLD?

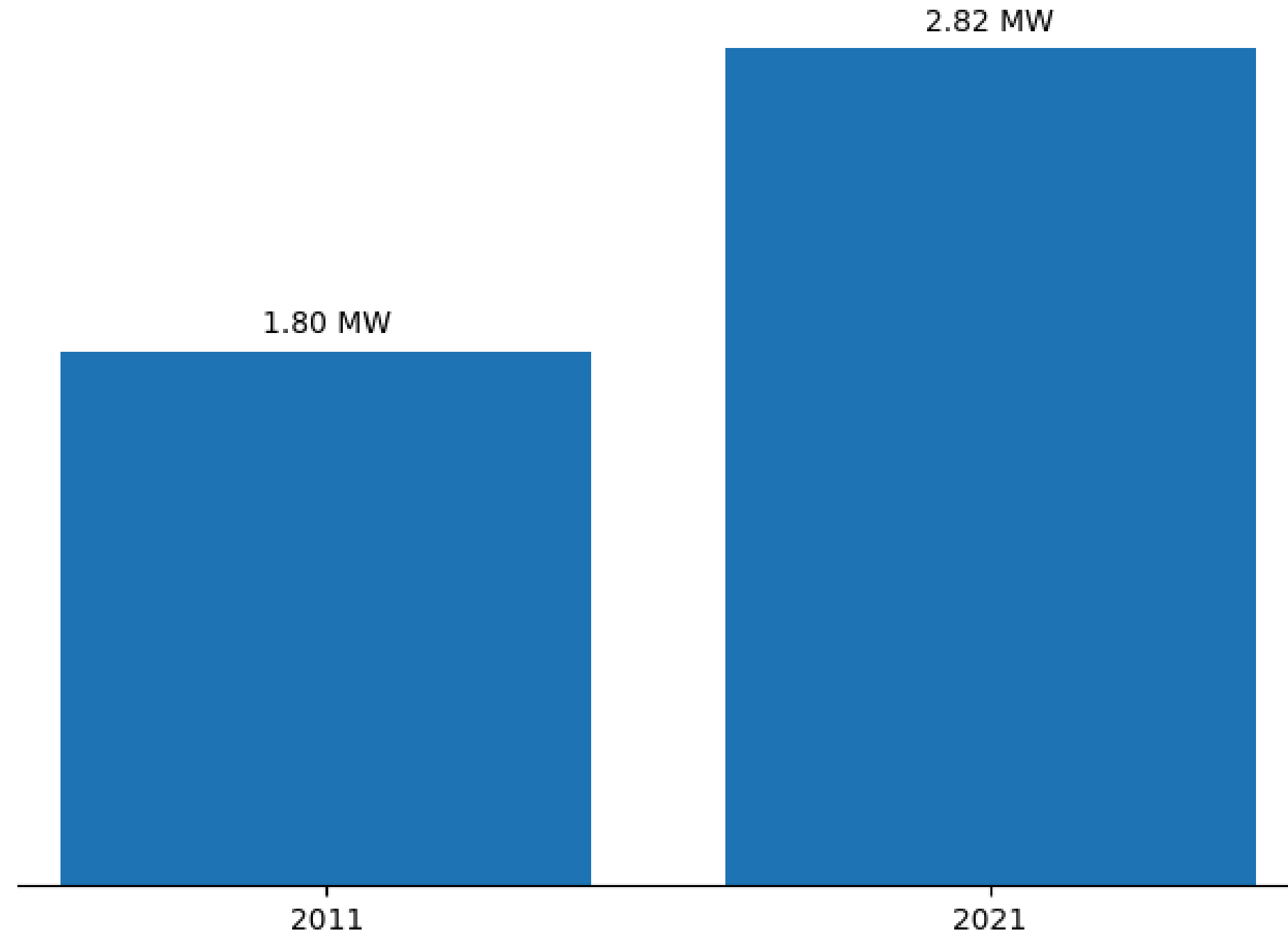


6,252 OF 70,195 TURBINES WERE AT LEAST 20 YEARS OLD AS OF 2022.

TURBINE RATED CAPACITY HAS INCREASED OVER TIME

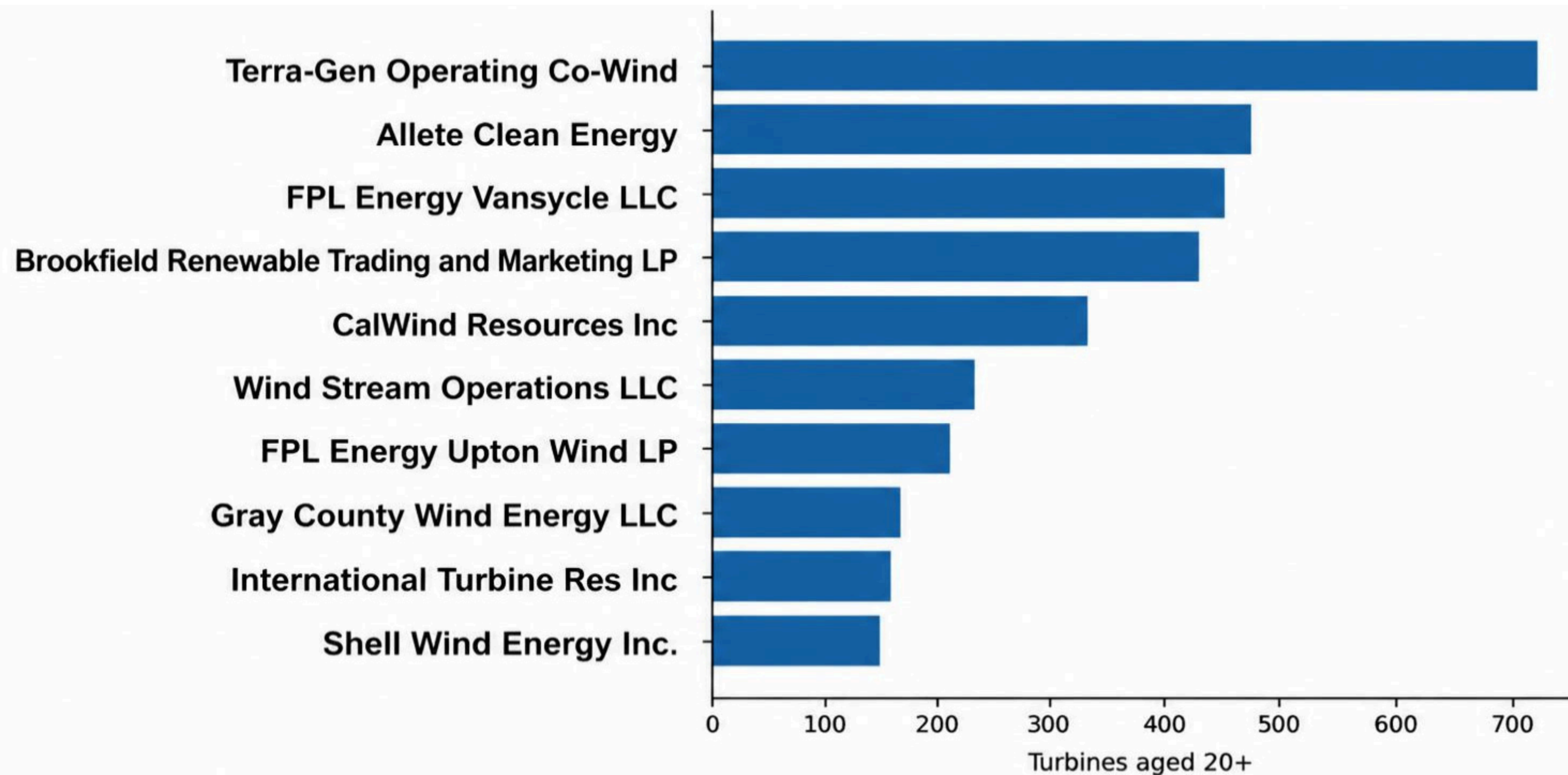


MODERN VS OLDER TURBINES



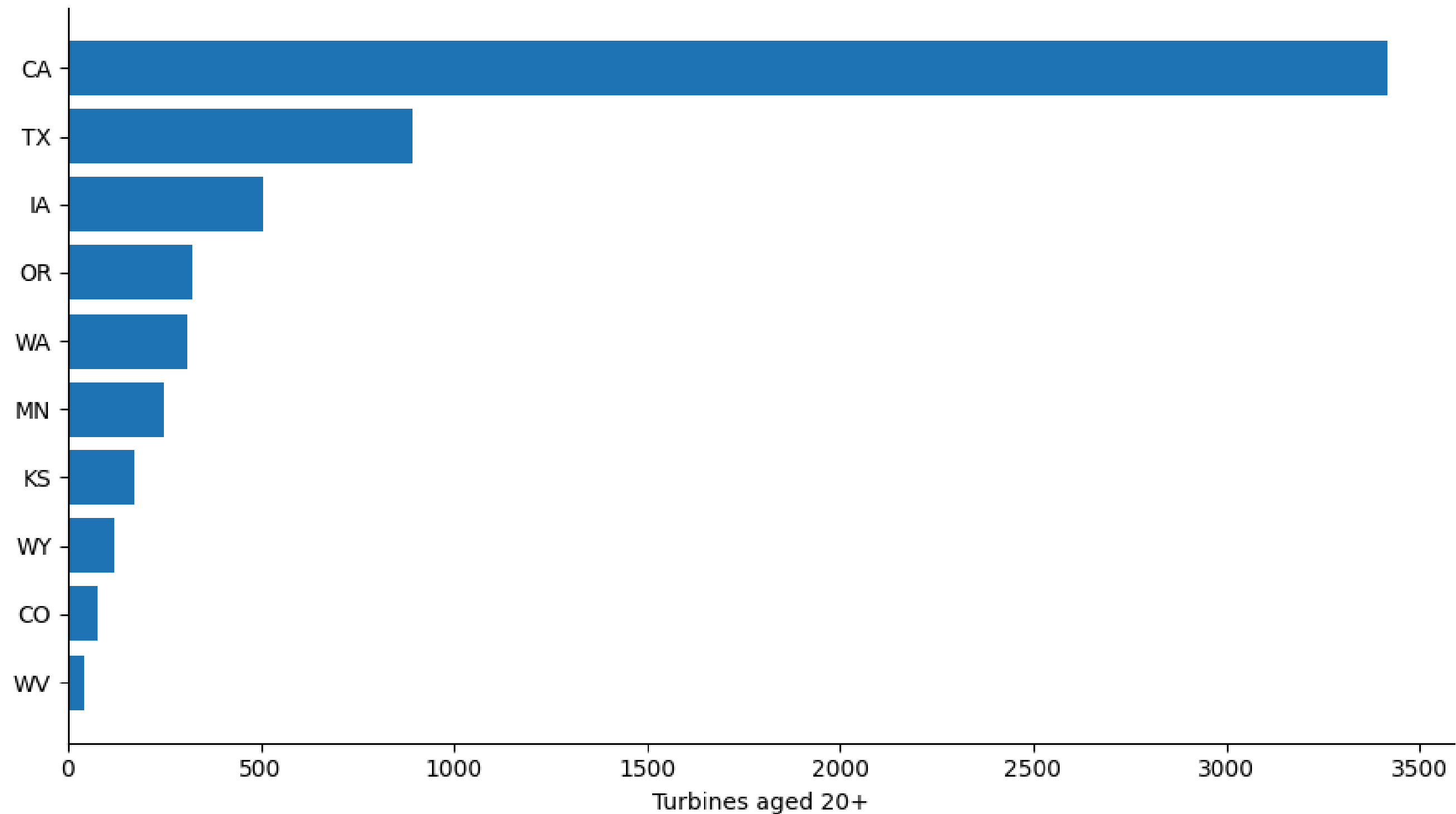
MEDIAN RATED CAPACITY INCREASED FROM 1.80 MW IN 2011 TO 2.82 MW IN 2021.

OPERATORS WITH THE LARGEST AGING TURBINE FLEETS



WHERE ARE AGING TURBINES CONCENTRATED?

States with the Largest Aging Turbine Fleets



RECOMMENDATIONS

Geographic Focus: Prioritize **California** (3,417 aging turbines), followed by **Texas** (894 aging turbines, 927 MW).

Operator Outreach: Start with **Terra-Gen**, followed by **Allete Clean Energy** and **FPL Energy Vansycle**.

Repowering Assessment: Evaluate replacing aging turbines with newer, higher capacity models.

THANK YOU



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